

Zaiyi Zheng

Ph.D. Student

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Summary

Ph.D. student in Computer Engineering at the University of Virginia, advised by Professor Jundong Li. My research interests lie in recommendation systems, retrieval-augmented generation, and large language models.

Education

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| Ph.D. University of Virginia , Computer Engineering
• Advisor: Prof. Jundong Li . | Charlottesville, VA, United States
Aug 2023 – present |
| B.E. University of Science and Technology of China , Computer Science | Hefei, Anhui, China
Sept 2018 – June 2022 |

Industry Experience

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|---|---|
| Meta , Research Advisor II
Topic: LLM for Recommendation. Adviser: Dr. Cassie Dong.
• Built a large-scale transformer-based model for effective and efficient ads ranking. | Bellevue, WA, United States
May 2026 – May 2027 |
| Nokia America , Machine Learning Research Intern
Topic: Generative Recommendation. Adviser: Dr. Liang Wu.
• Studied semantic ID (SID) based generative recommendation with large language models, focusing on reasoning-aware generation and efficient inference.
• Constructed explicit Chain-of-Thought (CoT) reasoning traces for each meta token in SID generation and trained a teacher model with a two-stage SFT-then-RL pipeline.
• Introduced auxiliary soft-thinking tokens for approximately 1B-parameter student models and distilled the teacher's CoT into continuous latent embeddings.
• Deployed only the student model at inference time, achieving comparable ranking quality to explicit CoT-based generation with reduced latency and generation cost. | Sunnyvale, CA, United States
Jan 2026 – May 2026 |
| Snap Inc. , Machine Learning Research Intern
Topic: Agentic Recommendation Systems. Adviser: Dr. Clark Ju.
• Developed a personalized LLM-based agent for multi-channel item recall by extending Hugging Face's TRL framework.
• Designed and implemented a soft-token mechanism to structurally generate routing weights over recall channels.
• Modified the vLLM inference engine to support high-throughput soft-token decoding and agent-style online interaction.
• Adapted Group Relative Policy Optimization (GRPO) to optimize adaptive routing over multiple recall models. | Bellevue, WA, United States
Sept 2025 – Dec 2025 |
| Snap Inc. , Research Assistant
Topic: Cold-Start Item Recommendation. Adviser: Dr. Neil Shah.
• Developed an adaptive cold-start item recommendation system for rapidly emerging items with sparse or zero interaction histories.
• Proposed MI4Rec, a PLM-based cold-start recommender that infers item representations via meta-item embedding composition.
• Evaluated on Amazon-Beauty, Toys, Sports, and Yelp under cold-start, warm-start, and general settings, achieving up to 15.71% relative improvement in Recall. | Bellevue, WA, United States
Dec 2024 – May 2025 |

Zerozero Robotics, Machine Learning Engineer Intern

Hangzhou, Zhejiang, China

Topic: Trajectory Planning. Adviser: Hao Zhang.

Mar 2022 – Aug 2022

- Built an outdoor trajectory planning system for physical quadrotors and constructed realistic evaluation environments in Unity.
- Reproduced an imitation learning framework and incorporated data aggregation to improve obstacle avoidance.
- Achieved an 18.2% relative reduction in navigation failure rate in outdoor environments.

Academic Experience

University of Virginia, Research Assistant

Charlottesville, VA, United States

Topic: Retrieval-Augmented Generation (RAG). Adviser: Prof. Jundong Li.

Aug 2023 – present

- Proposed CoRAG, a hybrid RAG framework that dynamically selects between global textual retrieval and graph-based relational retrieval.
- Designed a Cooperative-Retrievers (CoR) mixture-of-retrievers module with hierarchical gating to adaptively fuse heterogeneous retrieval signals.
- Demonstrated strong performance on STaRK semi-structured KGQA benchmarks, achieving up to 11.4% relative Hit@1 improvement over prior RAG baselines.

University of Texas at Austin, Research Assistant

Austin, TX, United States

Topic: Indoor Navigation. Adviser: Dr. Wuyang Chen.

May 2021 – Sept 2021

- Developed an efficient and transferable robot navigation method for sim-to-real deployment.
- Implemented a PPO-based navigation pipeline on Meta's Habitat platform.
- Proposed a contrastive learning component for visual representation learning with positive and negative sampling.

Interests

Research Areas: recommendation systems, retrieval-augmented generation, large language models, graph learning

Skills

Field of Study: Recommendation System, Retrieval Augmented Generation

Programming and Tools: Python, PyTorch, Pandas, C++, Unity, LaTeX, Git, Linux

Awards

Student Travel Award

2025

CIKM 2025 and BigData 2024.

Silver Scholarship for Outstanding Students

2019

University of Science and Technology of China, 2019 - 2022.

Talents Class Scholarship

2019

University of Science and Technology of China.